

todo Co-Formation Games

Jens Friedel

*School of Engineering Mathematics
and Technology
University of Bristol
Bristol, United Kingdom
jens.friedel@bristol.ac.uk*

Richard Pettigrew

*Department of Philosophy
University of Bristol
Bristol, United Kingdom
richard.pettigrew@bristol.ac.uk*

Martin Bullinger

*School of Engineering Mathematics
and Technology
University of Bristol
Bristol, United Kingdom
martin.bullinger@bristol.ac.uk*

Abstract—todo

Index Terms—Collective dynamics, Game theory, Network formation games, Network theory (graphs)

$$f_i^c(a) = -\alpha_c \|a_i\| - \frac{1}{\|k\|} \sum_{j \in K_i} |x_i - x_j| \quad (3)$$

b) *Algorithm:*

I. INTRODUCTION

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A. *Paper overview*

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II. METHODS

A. *Definitions & Equations*

TABLE I
LIST OF VARIABLES USED IN THIS DOCUMENT

Variable	Symbol	Domain
Number of nodes	n	$\mathbb{N}$
Node of concern	i	$\{1, \dots, n\}$
Set of i 's neighbours	K_i	$\mathbb{N}^{ k_i }$
Network graph ¹	G	$\{0, 1\}^{n \times n}$
Opinion	x_i	$\mathbb{R}$
Action profile	a_i	$\{0, 1\}^n$
Costs per edge	α_c	$[0, \infty[$
Pairwise distance	d_{ij}	$\mathbb{N}$
Network valuation	v	$] -\infty, 0]$
Network expenses	f^c	$] -\infty, 0[$
Cost function	c	$] -\infty, 0[$
Communication rate	r_c	$[0, \infty[$
Formation rate	r_f	$\{0, 1\}$
Convergence threshold	ε	$[0, \alpha_c]$

where a norm $\|\cdot\|$ is the L0 norm, returning the count of non-zero elements.

a) *Equations:*

- Cost function

$$c_i = v_i(a) + f^c(a) \quad (1)$$

- Valuation function

$$v_i(a) = - \sum_{j=1, \dots, n} d_{ij} \quad (2)$$

- Network expenses:

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1 while  $t < t_{\max}$ 
2   if  $(\text{mod}(t, r_c) = 0 \wedge \varepsilon < \Delta c)$  then
3     for  $i \in \{1, \dots, n\}$  do
4       | get opinion update  $\Delta x$ 
5     end for
6      $x \leftarrow x + \Delta x$ 
7   end if
8   if  $r_f = 1$  then
9     for  $i \in \{1, \dots, n\}$  do
10      | get action update  $\Delta a$ 
11    end for
12     $a \leftarrow a + \Delta a$ 
13  end if
14  Get new costs  $c_{t+1}$ 
15   $\Delta c = |c_{t+1} - c_t|$ 
16   $c_t \leftarrow c_{t+1}$ 
17   $t \leftarrow t + 1$ 
18 end while

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Algorithm 1. Dynamics of our synchronous move game

III. OLD STUFF FROM THE TEMPLATE



Fig. 1. A circle representing the Sun.

In Fig. 1 you can see a common representation of the Sun, which is a star that is located at the center of the solar system.

In Table II, you see the planets of the solar system and their average distance from the Sun. The distances were calculated with (1) that we presented in Section II.

¹The graph is represented as an adjacency matrix.

TABLE II
THE PLANETS OF THE SOLAR SYSTEM AND THEIR AVERAGE DISTANCE FROM
THE SUN

Planet	Distance (million km)
Mercury	57.9
Venus	108.2
Earth	149.6
Mars	227.9
Jupiter	778.6
Saturn	1,433.5
Uranus	2,872.5
Neptune	4,495.1

REFERENCES